# A transitive-side support deficit at a unique crossing

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ be transitive, $n \ge 12$, with a string C-marking and every maximal omission intransitive. Suppose the unique eligible $i$-edge is $\{a, b\}$, where $0 < i < r-1$ and $A, B$ are the two $G_i$-orbits. Assume $L = G_{<i}$ is transitive on $A$ and fixes $B$ pointwise, and that some generator of index greater than $i$ acts nontrivially on both orbits. These are precisely the transitive-left-side hypotheses used below; they do not assume an intersection property for a projected group.

**Lemma 1** (An even centralizer on paired rows)**.** *Let $p \ge 4$ be even. A group $H$ preserves $p$ disjoint pairs and induces the full natural $S_p$ on those pairs. Every involution centralizing $H$ on their $2p$ points is even.*

*Proof.* There are either one or two $H$-orbits, because an orbit meets all the pairs in a constant positive number of points. If $H$ is transitive, a nonidentity centralizing involution has no fixed point: commutation transports any fixed point to its entire orbit. It is therefore a product of $p$ transpositions, which is even. If there are two orbits, each meets every pair once. The natural action on each orbit is $S_p$ and has trivial centralizer. A commuting permutation either preserves both orbits, in which case it is the identity, or exchanges them. In the latter case a commuting involution again consists of $p$ transpositions. This proves the claim in both cases. □

**Proposition 2** (Transitive-side support and strict endpoint exclusion)**.** *The largest shared label is $h = i + 1$. If $h = r - 1$, then $r \le \lfloor (n-1)/2 \rfloor$. If $h < r - 1$, put $X = \operatorname{supp}(G_{>h})$. Then*

$$X \subseteq B \setminus \{b\}, \qquad |\operatorname{Fix}(G_{>h})| \ge |A| + 1 \ge 2h + 1, \qquad h \le \frac{n - |X| - 1}{2}.$$

*At equality in the last bound, $|A| = 2h$ and the fixed set is exactly $A \cup \{b\}$.*

*Proof.* Each generator of index greater than $i$ centralizes $L$. Its restriction to $A$ is therefore either the identity or fixed-point-free. But every generator of index at least $i + 2$ fixes $a$, by the unique-crossing lemma. Hence only $\tau = \rho_{i+1}$ can act nontrivially on $A$, and it does. It is fixed-point-free there, so write $|A| = 2m$. The accepted commuting-tail rank lemma, applied in the dual index order to $L$ and $\tau$, gives

$$i \le m - 1. \tag{1}$$

Its hypotheses hold literally: the generators of $L$ are supported on $A$, its action is transitive, the earlier-in-the-dual-order involution $\tau$ centralizes it and moves $A$, and all global omissions are intransitive.

When $h < r - 1$, the tail fixes $A$ pointwise by the preceding centralizer argument. It also fixes $b$, since every tail index is at least $i + 2$. Thus it fixes at least $2m + 1$ points, and (1) gives $2m + 1 \ge 2i + 3 = 2h + 1$. The equality description follows directly from these two inequalities and inclusion of $A \cup \{b\}$ in the fixed set.

It remains to exclude rank failure when $h = r - 1$. On $B$, the whole $G_i$-action is then generated by $\tau$, since $L$ fixes $B$. Hence $B$ consists of two points $b, b'$. Consequently $n = 2m + 2$ and $r = i + 2 \le m + 1$. A failure of the asserted rank bound would force

$$i = m - 1, \qquad r = m + 1. \tag{2}$$

The involution $\tau$ exchanges $m$ pairs on $A$ and the pair $B$. Its evenness forces $m$ odd; moreover $n \geq 12$ gives $m \geq 5$.

Consider the action of $L$ on the $m$ pairs of $\tau$ in $A$. The proof of the accepted commuting-tail lemma shows that *every* distinguished generator of $L$ is essential in this block action: omitting a dispensable generator, and using $\tau$ to join its two possible point orbits, would make the corresponding global omission transitive. Their $m-1$ block images thus form an independent involutory string. Whiston's equality theorem identifies the block image as $S_m$. The accepted maximal symmetric-string classification now makes this marking an ordinary adjacent- transposition path: degree five is covered by its complete finite certificate, and odd degrees at least seven by its uniform theorem. The degree-six outer action cannot occur. Neither step assumes projected IP.

Put $H = G_{<i-1}$, the omission of the near endpoint of $L$. Its pair action is $S_{m-1}$ on all but the anchor pair $\{a, a'\}$, where $a' = a^\tau$. Every member of $H$ fixes $a$ by the unique crossing, and fixes $a'$ by commutation with $\tau$. Its only fixed points in $A$ are these two, since its block action moves every other pair.

The crossing involution $s = \rho_i$ has the form

$$s = (a\ b)u,$$

where $u$ is an involution on $A \setminus \{a\}$; the other point $b'$ is fixed. This follows because $(a\ b)$ is the only cross-orbit transposition and no other point remains in $B$ on which $s$ could act. String commutation makes $u$ centralize $H$ on $A$. Since $u$ fixes $a$, it also fixes the other $H$-fixed point $a'$. On the remaining $2(m-1)$ points, Lemma 1 applies with $p = m-1$, which is even and at least four. Thus $u$ is even. The permutation $(a\ b)u$ is odd, contradicting $G \leq A_n$. This excludes (2) and completes the proof. □

**Scope and attribution.** This proves the numerical support conclusion and the strict endpoint exclusion used in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for $A_n$*, arXiv:1605.09173v1, Proposition 5.22 (physical page 31), with its standing hypotheses explicit. The exact equality fixed-set description above does not claim the source's complete coloured-graph classification. The proof reuses the accepted unique-crossing and commuting-tail lemmas and all paid L047/Whiston maximal-string inputs, including the whole degree-five certificate. No new computation is a premise; there is no new formal declaration, whole-original admission or question credit. The intransitive left-side branch of Proposition 5.23 and the remaining full-IP square/attachment rigidity are separate obligations.